SOWJANYA SURADA

AI / Machine Learning Engineer — Fresher

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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech, 2026) with focus on Computer Vision and Machine Learning, hands-on experience building and deploying real-time detection and prediction systems using CNNs, YOLO, and TensorFlow/Keras. Led development of a self-driving car simulation using behavioral cloning, and built a drone detection model achieving 85%+ accuracy. Strong foundation in Python, deep learning, and end-to-end ML workflows from data collection to deployment.

TECHNICAL SKILLS

- Languages: Python, Java
- ML/DL Frameworks: TensorFlow, Keras, YOLOv5, YOLOv11n, OpenCV, MediaPipe, PyTorch, Scikit-learn, CNN, NumPy, Pandas
- Deployment & Tools: Streamlit, Flask, SocketIO, Tkinter, Teachable Machine, Roboflow, Git, Jupyter, Google Colab
- Databases: MySQL, Oracle SQL

ACADEMIC PROJECT — FINAL YEAR

Self-Driving Car Simulation using Deep Learning | *B.Tech Capstone, Team of 4* Dec 2025 – Apr 2026

- Built an end-to-end autonomous driving pipeline using the Udacity Self-Driving Car Simulator, applying behavioral cloning to map camera images directly to steering angles
- Designed and trained an NVIDIA-style CNN (TensorFlow/Keras) on preprocessed and augmented driving data (cropping, resizing, normalization, flipping) to predict continuous steering commands
- Built the real-time inference pipeline with Flask + SocketIO to stream simulator camera frames to the trained model and return steering/throttle commands
- Achieved successful autonomous driving on straight roads and moderate curves in the simulator with no human intervention

PERSONAL PROJECTS

Drone Detection using YOLOv11n | *Personal Project*

- Built a real-time object detection model for drones using annotated Roboflow datasets, achieving 85%+ accuracy while staying lightweight for real-time use
- GitHub: github.com/sowjanya-py/drone-detection

Plant Disease Detection App | *Personal Project*

- Trained a YOLOv5 model on a Roboflow plant-disease dataset; built a Streamlit UI showing bounding boxes, confidence scores, and treatment suggestions
- GitHub: github.com/sowjanya-py/plant-disease-detection | Live demo available

AIVA (Telegram Bot) · Face Landmark Detection · Python GUI Tools | *Personal Projects*

- AI assistant bot for conversational task automation; real-time facial landmark detection using MediaPipe; Tkinter utilities and a web-based student report card generator

EXPERIENCE

AI & Computer Vision Intern | *AIMER Society* May – Jul 2025

- Developed CNN and YOLO-based models for image classification and detection tasks
- Built a sign language detector and multiple image classifiers

DSA Intern | *CodeBeat* Jun – Aug 2024

- Solved data structures and algorithms problems, strengthening core problem-solving skills

Python Development Intern | *Technohacks EduTech* May – Jun 2024

- Built Python automation tools and GUI applications using Tkinter

EDUCATION

B.Tech, Computer Science Engineering — CGPA: 8.0 | *Miracle Educational Society Group of Institutions, JNTU-GV* 2022 – 2026

Senior Secondary (XII), CBSE — 80% | *Kendriya Vidyalaya, INS Kalinga, Visakhapatnam* 2022

Secondary (X), CBSE — CGPA: 8.11 | *Kendriya Vidyalaya, INS Kalinga, Visakhapatnam* 2020

CERTIFICATIONS & ACTIVITIES

- Data Science with Python — Teachnook (in association with IIT Bhubaneswar)
- Tech Fest Lead; Hackathon participant | Languages: Telugu, Hindi, English